



**NORTH
SYDNEY
GIRLS HIGH
SCHOOL**

2024 HSC ASSESSMENT TASK 3

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total Marks – 38

Section I – 4 marks (pages 2–3)

- Attempt Questions 1–4
- Allow about 8 minutes for this section

Section II – 34 marks (pages 5–7)

- Attempt Questions 5–7
- Allow about 52 minutes for this section

NAME: _____ STUDENT NUMBER: _____

TEACHER: _____

	Multiple Choice	Question 5	Question 6	Question 7	Total
Proof	/3	/ 2	/7	/4	/16
Integration	/1	/ 8	/5	/8	/22
Total	/4	/ 10	/12	/12	/38

Section I

4 marks

Attempt Questions 1–4

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–4.

- 1 Without evaluating the integrals, which one of the following will give an answer of zero?

A. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\cos^3 \theta + 1}{\cos^2 \theta} dx$

B. $\int_{-1}^1 (x^2 - 1)^2 dx$

C. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x \cos x dx$

D. $\int_{-2}^2 |x^2 - 4| dx$

- 2 Given that $x > 0$, what is the smallest value of $\frac{x}{2} + \frac{8}{x}$?

A. 16

B. 8

C. 4

D. 2

3 If a and b are positive, which of the following is false?

A. $(a + b)^2 \geq 4ab$

B. $\frac{1}{a} + \frac{1}{b} < \frac{4}{a + b}$

C. $\frac{a}{b} + \frac{b}{a} \geq 2$

D. $(a + b)^2 \leq 2(a^2 + b^2)$

4 Which of the following inequalities is incorrect?

A. $\int_0^1 \ln(1 + x) \, dx > \int_0^1 \frac{x - 1}{e - 1} \, dx$

B. $\int_0^{\frac{\pi}{4}} \sin x \, dx > \int_0^{\frac{\pi}{4}} \sin^2 x \, dx$

C. $\int_0^1 e^x \, dx > \int_0^1 \frac{1}{1 + x^2} \, dx$

D. $\int_0^1 e^{-x^2} \, dx > \int_0^1 e^{-x^3} \, dx$

End of Section I

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Section II

34 marks

Attempt Questions 5–7

Allow about 52 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 5 (10 marks) Use a SEPARATE Writing Booklet

- (a) Prove for any real numbers x , y , z and u that **2**

$$(x^2 + y^2)(z^2 + u^2) \geq (xz + yu)^2.$$

- (b) Using integration by parts find $\int \sqrt{x} \ln x \, dx$. **2**

- (c) By writing $\frac{9x}{(x+2)(x-1)^2}$ in the form $\frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$, **4**

find $\int \frac{9x}{(x+2)(x-1)^2} \, dx$.

- (d) Find $\int \frac{1}{(1+x)\sqrt{x}} \, dx$. **2**

Question 6 (12 marks) Use a SEPARATE Writing Booklet

- (a) You are given that for positive real numbers $x_1, x_2, \dots, x_n$

$$\frac{x_1 + x_2 + \dots + x_n}{n} \geq \sqrt[n]{x_1 x_2 \dots x_n}.$$

- (i) Show that for positive real numbers x, y and z

1

$$\frac{x^3}{yz} + y + z \geq 3x.$$

- (ii) Hence, or otherwise show for positive real numbers x, y and z

2

$$\frac{x^3}{yz} + \frac{y^3}{xz} + \frac{z^3}{xy} \geq x + y + z.$$

- (b) Use symmetry to evaluate $\int_0^\pi \frac{\sin 4x}{1 + \sin x} dx$.

2

- (c) Use the substitution $t = \tan \frac{x}{2}$ to find

3

$$\int_0^{\frac{2\pi}{3}} \frac{dx}{5 + 3 \sin x + 4 \cos x}.$$

- (d) (i) Show that $\sin\left(\theta + \frac{\pi}{2}\right) = \cos \theta$.

1

- (ii) Consider $f(x) = \sin(ax)$ where a is a constant.

3

Prove by mathematical induction that $f^{(n)}(x) = a^n \sin\left(ax + \frac{n\pi}{2}\right)$

where $n \in \mathbb{Z}^+$ and $f^{(n)}(x)$ represents the n^{th} derivative of $f(x)$.

Question 7 (12 marks) Use a SEPARATE Writing Booklet

(a) Use a suitable substitution to find $\int \frac{\sqrt{1+x^2}}{x^4} dx$. **3**

(b) Let $I_n = \int_0^1 x^n \sqrt{1-x^3} dx$ for $n \geq 2$.

(i) Show that for $n \geq 5$ **3**

$$I_n = \frac{2n-4}{2n+5} I_{n-3}.$$

(ii) Hence find I_5 . **2**

(c) Prove by mathematical induction for $n \geq 2$: **4**

$$n \left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2n-1} \right) > (n+1) \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2n} \right).$$

You may use the following result where $k \geq 2$:

$$\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1} \right) - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2k} \right) > \frac{k}{(2k+1)(2k+2)}$$

End of paper

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Q1) C Q2) C Q3) B Q4) D

Question 5

a) Consider

$$\begin{aligned}
 & (x^2+y^2)(z^2+u^2) - (xz+yu)^2 \\
 &= x^2z^2 + x^2u^2 + y^2z^2 + y^2u^2 - (x^2z^2 + 2xzyu + y^2u^2) \\
 &= \cancel{x^2z^2} + x^2u^2 + y^2z^2 + \cancel{y^2u^2} - \cancel{x^2z^2} - 2xzyu - \cancel{y^2u^2} \\
 &= x^2u^2 + y^2z^2 - 2xzyu \\
 &= (xu - yz)^2 \geq 0 \text{ as } (xu - yz)^2 \text{ is a perfect square} \\
 &\therefore (x^2+y^2)(z^2+u^2) - (xz+yu)^2 \geq 0 \\
 &\therefore (x^2+y^2)(z^2+u^2) \geq (xz+yu)^2
 \end{aligned}$$

b) $\int \sqrt{x} \ln x \, dx$

$$\begin{aligned}
 u &= \ln x & v' &= \sqrt{x} \\
 u' &= \frac{1}{x} & v &= \frac{2x^{\frac{3}{2}}}{3}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2}{3} x^{\frac{3}{2}} \ln x - \int \frac{1}{x} \times \frac{2}{3} x^{\frac{3}{2}} \, dx \quad (\text{by parts}) \\
 &= \frac{2}{3} x^{\frac{3}{2}} \ln x - \frac{2}{3} \int x^{\frac{1}{2}} \, dx \\
 &= \frac{2}{3} x^{\frac{3}{2}} \ln x - \frac{2}{3} \times \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \\
 &= \frac{2}{3} x^{\frac{3}{2}} \ln x - \frac{4}{9} x^{\frac{3}{2}} + C
 \end{aligned}$$

$$c) \frac{9x}{(x+2)(x-1)^2} \equiv \frac{A}{x+2} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

$$9x = A(x-1)^2 + B(x+2)(x-1) + C(x+2)$$

$$\text{sub } x=1: 9 = 3C \therefore C=3$$

$$\text{sub } x=-2: -18 = A(-3)^2$$

$$-18 = 9A$$

$$\therefore A = -2$$

$$\text{coefficient of } x^2: 0 = A + B$$

$$0 = -2 + B$$

$$B = 2$$

$$\therefore \frac{9x}{(x+2)(x-1)^2} \equiv \frac{-2}{x+2} + \frac{2}{x-1} + \frac{3}{(x-1)^2}$$

$$\begin{aligned}
 \therefore \int \frac{9x}{(x+2)(x-1)^2} \, dx &= \int \left(\frac{-2}{x+2} + \frac{2}{x-1} + \frac{3}{(x-1)^2} \right) dx \\
 &= -2 \ln|x+2| + 2 \ln|x-1| - \frac{3}{(x-1)} + C
 \end{aligned}$$

$$\begin{aligned}
 d) \quad & \int \frac{1}{(1+x)\sqrt{x}} dx && \text{let } u = \sqrt{x} \\
 & = 2 \int \frac{1}{(1+u^2)} \cdot \frac{du}{2\sqrt{x}} && \frac{du}{dx} = \frac{1}{2\sqrt{x}} \\
 & = 2 \int \frac{1}{1+u^2} du && du = \frac{dx}{2\sqrt{x}} \\
 & = 2 \tan^{-1} u \\
 & = 2 \tan^{-1}(\sqrt{x}) + C
 \end{aligned}$$

Question 6

a) i) $\frac{1}{3} \left(\frac{x^3}{yz} + y + z \right) \geq \sqrt[3]{\frac{x^3}{yz} \cdot y \cdot z}$ using AM-GM given

$$\frac{x^3}{yz} + y + z \geq 3 \sqrt[3]{x^3}$$

$$\frac{x^3}{yz} + y + z \geq 3x \quad \text{--- (1)}$$

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ii) Similarly

$$\frac{y^3}{xz} + x + z \geq 3y \quad \text{--- (2)} \quad \text{and} \quad \frac{z^3}{xy} + x + y \geq 3z \quad \text{--- (3)}$$

$$(1) + (2) + (3)$$

$$\frac{x^3}{yz} + y + z + \frac{y^3}{xz} + x + z + \frac{z^3}{xy} + x + y \geq 3x + 3y + 3z$$

$$\frac{x^3}{yz} + \frac{y^3}{xz} + \frac{z^3}{xy} + 2(x + y + z) \geq 3(x + y + z)$$

$$\therefore \frac{x^3}{yz} + \frac{y^3}{xz} + \frac{z^3}{xy} \geq 3(x + y + z) - 2(x + y + z)$$

$$\therefore \frac{x^3}{yz} + \frac{y^3}{xz} + \frac{z^3}{xy} \geq x + y + z$$

b)
$$\begin{aligned} I &= \int_0^{\pi} \frac{\sin 4x}{1 + \sin x} dx \\ &= \int_0^{\pi} \frac{\sin 4(\pi - x)}{1 + \sin(\pi - x)} dx \\ &= \int_0^{\pi} \frac{\sin(4\pi - 4x)}{1 + \sin x} dx \\ &= \int_0^{\pi} \frac{\sin(-4x)}{1 + \sin x} dx \\ &= \int_0^{\pi} \frac{-\sin 4x}{1 + \sin x} dx \quad \text{as sine is odd} \\ &= -I \end{aligned}$$

$$2I = 0$$

$$I = 0$$

$$\therefore \int_0^{\pi} \frac{\sin 4x}{1 + \sin x} dx = 0$$

$$\begin{aligned}
 c) \quad & \int_0^{\frac{2\pi}{3}} \frac{dx}{5 + 3\sin x + 4\cos x} \\
 &= \int_0^{\sqrt{3}} \frac{1}{5 + \frac{3+2t}{1+t^2} + \frac{4(1-t^2)}{1+t^2}} \times \frac{2dt}{1+t^2} \\
 &= \int_0^{\sqrt{3}} \frac{2dt}{5(1+t^2) + 6t + 4(1-t^2)} \\
 &= \int_0^{\sqrt{3}} \frac{2dt}{5+5t^2+6t+4-4t^2} \\
 &= \int_0^{\sqrt{3}} \frac{2dt}{t^2+6t+9} \\
 &= 2 \int_0^{\sqrt{3}} \frac{dt}{(t+3)^2} \\
 &= 2 \left[\frac{(t+3)^{-1}}{-1} \right]_0^{\sqrt{3}} \\
 &= -2 \left[\frac{1}{t+3} \right]_0^{\sqrt{3}} \\
 &= -2 \left(\frac{1}{\sqrt{3}+3} - \frac{1}{3} \right) \\
 &= 2 \left(\frac{1}{3} - \frac{1}{3+\sqrt{3}} \right) \\
 &= 2 \left(\frac{1}{3} - \frac{3-\sqrt{3}}{6} \right) \\
 &= 2 \times \frac{2-3+\sqrt{3}}{6} \\
 &= \frac{\sqrt{3}-1}{3}
 \end{aligned}$$

$$\text{let } t = \tan \frac{x}{2}$$

$$x = 2 \tan^{-1} t$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$x=0 \Rightarrow t=0$$

$$x = \frac{2\pi}{3} \Rightarrow t = \sqrt{3}$$

d) i) $\sin\left(\theta + \frac{\pi}{2}\right)$

$$= \sin\theta \cos\frac{\pi}{2} + \sin\frac{\pi}{2} \cos\theta$$

$$= \sin\theta \times 0 + 1 \times \cos\theta$$

$$= \cos\theta$$

ii) Prove true for $n=1$:

$$f(x) = \sin(ax)$$

$$\text{LHS} = f'(x) \quad \text{RHS} = a' \sin\left(ax + \frac{\pi}{2}\right)$$

$$= a \cos(ax)$$

$$= a \sin\left(ax + \frac{\pi}{2}\right)$$

$$= a \cos(ax) \quad (\text{using part (i)})$$

Assume result is true for $n=k$, $k \in \mathbb{Z}^+$

$$\therefore f^{(k)}(x) = a^k \sin\left(ax + \frac{k\pi}{2}\right)$$

Prove result is true for $n=k+1$

$$\text{RTP: } f^{(k+1)}(x) = a^{k+1} \sin\left(ax + \frac{(k+1)\pi}{2}\right)$$

From the assumption:

$$f^{(k)}(x) = a^k \sin\left(ax + \frac{k\pi}{2}\right)$$

$$f^{(k+1)}(x) = \frac{d}{dx} (f^{(k)}(x))$$

$$= \frac{d}{dx} \left(a^k \sin\left(ax + \frac{k\pi}{2}\right) \right)$$

$$= a^k \times a \cos\left(ax + \frac{k\pi}{2}\right)$$

$$= a^{k+1} \cos\left(ax + \frac{k\pi}{2}\right)$$

$$= a^{k+1} \sin\left(\left(ax + \frac{k\pi}{2}\right) + \frac{\pi}{2}\right) \quad \text{using part (i) where } \theta = ax + \frac{k\pi}{2}$$

$$= a^{k+1} \sin\left(ax + \frac{k\pi}{2} + \frac{\pi}{2}\right)$$

$$= a^{k+1} \sin\left(ax + \frac{(k+1)\pi}{2}\right)$$

$$= \text{RHS.}$$

$\therefore$ The result is true for $n \in \mathbb{Z}^+$ by the principle of mathematical induction.

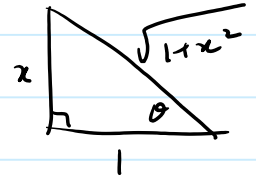
Question 7

$$\begin{aligned}
 a) \quad & \int \frac{\sqrt{1+x^2}}{x^4} dx \\
 &= \int \frac{\sqrt{1+\tan^2 \theta}}{\tan^4 \theta} \sec^2 \theta d\theta \\
 &= \int \frac{\sec \theta}{\tan^4 \theta} \sec^2 \theta d\theta \\
 &= \int \sec^3 \theta \times \frac{\cos^4 \theta}{\sin^4 \theta} d\theta \\
 &= \int \frac{1}{\cos^3 \theta} \times \frac{\cos^4 \theta}{\sin^4 \theta} d\theta \\
 &= \int \cos \theta (\sin \theta)^{-4} d\theta \\
 &= \frac{(\sin \theta)^{-3}}{-3} \\
 &= -\frac{1}{\sin^3 \theta}
 \end{aligned}$$

$$\text{let } x = \tan \theta$$

$$\frac{dx}{d\theta} = \sec^2 \theta$$

$$dx = \sec^2 \theta d\theta$$



$$\sin \theta = \frac{x}{\sqrt{1+x^2}}$$

$$\begin{aligned}
 &= -\frac{1}{\left(\frac{x}{\sqrt{1+x^2}}\right)^3} \\
 &= -\frac{(1+x^2)\sqrt{1+x^2}}{x^3} + C
 \end{aligned}$$

$$\begin{aligned}
 b) i) \quad I_n &= \int_0^1 x^n \sqrt{1-x^2} dx \quad n \geq 2 \quad u = x^{n-2} \quad v' = x^2 (1-x^2)^{\frac{1}{2}} \\
 &\quad u' = (n-2)x^{n-3} \quad v = -\frac{1}{3}(1-x^2)^{\frac{3}{2}} \times \frac{2}{3} \\
 &\quad = -\frac{2}{9}(1-x^2)^{\frac{3}{2}}
 \end{aligned}$$

$$\begin{aligned}
 I_n &= \left[x^{n-2} \times -\frac{2}{9}(1-x^2)^{\frac{3}{2}} \right]_0^1 - \int_0^1 (n-2)x^{n-3} \times -\frac{2}{9}(1-x^2)^{\frac{3}{2}} dx \quad (\text{by parts}) \\
 &= 0 - 0 + \frac{2}{9}(n-2) \int_0^1 x^{n-3} (1-x^2)^{\frac{3}{2}} dx \\
 &= \frac{2}{9}(n-2) \int_0^1 x^{n-3} (1-x^2) \sqrt{1-x^2} dx \\
 &= \frac{2}{9}(n-2) \int_0^1 x^{n-3} \sqrt{1-x^2} - x^n \sqrt{1-x^2} dx \\
 &= \frac{2}{9}(n-2) \left[\int_0^1 x^{n-3} \sqrt{1-x^2} dx - \int_0^1 x^n \sqrt{1-x^2} dx \right] \\
 &= \frac{2}{9}(n-2) [I_{n-3} - I_n] \\
 &= \frac{2}{9}(n-2) I_{n-3} - \frac{2}{9}(n-2) I_n
 \end{aligned}$$

$$I_n + \frac{2}{9}(n-2)I_n = \frac{2}{9}(n-2)I_{n-3}$$

$$9I_n + 2(n-2)I_n = 2(n-2)I_{n-3}$$

$$I_n(9+2n-4) = (2n-4)I_{n-3}$$

$$(2n+5)I_n = (2n-4)I_{n-3}$$

$$I_n = \frac{2n-4}{2n+5} I_{n-3}$$

$$\begin{aligned}
 ii) \quad I_2 &= \int_0^1 x^2 \sqrt{1-x^2} dx \quad I_5 = \frac{2(5)-4}{2(5)+5} I_2 \\
 &= -\frac{1}{3} \left[(1-x^2)^{\frac{3}{2}} \times \frac{2}{3} \right]_0^1 \\
 &= -\frac{2}{9} [0 - 1] \\
 &= \frac{2}{9} \\
 &= \frac{4}{45}
 \end{aligned}$$

c) Prove true for $n=2$

$$\begin{aligned} \text{LHS} &= 2\left(1 + \frac{1}{3}\right) & \text{RHS} &= (2+1)\left(\frac{1}{2} + \frac{1}{4}\right) \\ &= 2 \times \frac{4}{3} & &= 3 \times \frac{3}{4} \\ &= \frac{8}{3} & &= \frac{9}{4} \end{aligned}$$

$$\therefore \text{LHS} > \text{RHS}$$

$\therefore$ result is true for $n=2$

Assume result is true for $n=k$, $k \geq 2$

$$k\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1}\right) > (k+1)\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2k}\right)$$

Prove result is true for $n=k+1$

$$\text{RTP: } (k+1)\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1} + \frac{1}{2k+1}\right) > (k+1+1)\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2k} + \frac{1}{2k+2}\right)$$

$$\text{i.e. } (k+1)\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1} + \frac{1}{2k+1}\right) > (k+2)\left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2k} + \frac{1}{2k+2}\right)$$

From the given identity

$$\begin{aligned} \left(1 - \frac{1}{2}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \left(\frac{1}{5} - \frac{1}{6}\right) + \dots + \left(\frac{1}{2k-1} - \frac{1}{2k}\right) &> \frac{k}{(2k+1)(2k+2)} \quad \text{we can deduce} \\ 1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1} &> \frac{k}{(2k+1)(2k+2)} + \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots + \frac{1}{2k}\right) \quad (*) \end{aligned}$$

$$\begin{aligned} \text{LHS} &= (k+1)\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1} + \frac{1}{2k+1}\right) \\ &= k\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1}\right) + 1\left(1 + \frac{1}{3} + \frac{1}{5} + \dots + \frac{1}{2k-1}\right) + \frac{k+1}{2k+1} \\ &> \underbrace{(k+1)\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}\right)}_{\text{using the assumption}} + \underbrace{\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k} + \frac{k}{(2k+1)(2k+2)}\right)}_{\text{using } (*)} + \frac{k+1}{2k+1} \end{aligned}$$

$$= (k+2)\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}\right) + \frac{k}{(2k+1)(2k+2)} + \frac{k+1}{2k+1}$$

$$= (k+2)\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}\right) + \frac{k + (k+1)(2k+2)}{(2k+1)(2k+2)}$$

$$= (k+2)\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}\right) + \frac{k + 2k^2 + 2k + 2k + 2}{(2k+1)(2k+2)}$$

$$= (k+2)\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}\right) + \frac{2k^2 + 5k + 2}{(2k+1)(2k+2)}$$

$$= (k+2)\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k}\right) + \frac{(2k+1)(k+2)}{(2k+1)(2k+2)}$$

$$= (k+2)\left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2k} + \frac{1}{2k+2}\right)$$

$$= \text{RHS}$$

$\therefore$ The result is true by the principle of mathematical induction for $n \geq 2$